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In [1]: import pandas as pd
import matplotlib.pyplot as plt
```

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In [2]: #Task 1 - Load Dataset and Display First 5 Rows

sales_df = pd.read_csv("retail_sales.csv")
print(sales_df.head())
```

	product	category	date	sales_amount
0	Laptop	Electronics	2026-01-05	55000
1	T-Shirt	Clothing	2026-01-10	1200
2	Rice	Groceries	2026-01-15	2500
3	Mobile	Electronics	2026-02-05	30000
4	Jeans	Clothing	2026-02-12	2200

```
In [4]: sales_df
```

Out[4]:

	product	category	date	sales_amount
0	Laptop	Electronics	2026-01-05	55000
1	T-Shirt	Clothing	2026-01-10	1200
2	Rice	Groceries	2026-01-15	2500
3	Mobile	Electronics	2026-02-05	30000
4	Jeans	Clothing	2026-02-12	2200
5	Oil	Groceries	2026-02-20	1800
6	Headphones	Electronics	2026-03-08	5000
7	Shoes	Clothing	2026-03-15	3500
8	Flour	Groceries	2026-03-20	1200
9	Tablet	Electronics	2026-04-05	25000
10	Jacket	Clothing	2026-04-12	4500
11	Vegetables	Groceries	2026-04-18	1800
12	Laptop	Electronics	2026-05-05	60000
13	T-Shirt	Clothing	2026-05-10	1500
14	Rice	Groceries	2026-05-20	2800
15	Mobile	Electronics	2026-06-05	32000
16	Jeans	Clothing	2026-06-12	2400
17	Oil	Groceries	2026-06-20	2000
18	Headphones	Electronics	2026-07-08	5500
19	Shoes	Clothing	2026-07-15	3800
20	Flour	Groceries	2026-07-20	1400
21	Tablet	Electronics	2026-08-05	27000
22	Jacket	Clothing	2026-08-12	4800
23	Vegetables	Groceries	2026-08-18	1900
24	Laptop	Electronics	2026-09-05	58000
25	T-Shirt	Clothing	2026-09-10	1300
26	Rice	Groceries	2026-09-20	2600
27	Mobile	Electronics	2026-10-05	31000
28	Jeans	Clothing	2026-10-12	2300
29	Oil	Groceries	2026-10-20	2100
30	Headphones	Electronics	2026-11-08	5200
31	Shoes	Clothing	2026-11-15	3600
32	Flour	Groceries	2026-11-20	1300
33	Tablet	Electronics	2026-12-05	26000
34	Jacket	Clothing	2026-12-12	4600
35	Vegetables	Groceries	2026-12-18	2000

```
In [5]: #Task 2 - Bar Chart: Total Sales by Category

category_sales = sales_df.groupby("category")["sales_amount"].sum()

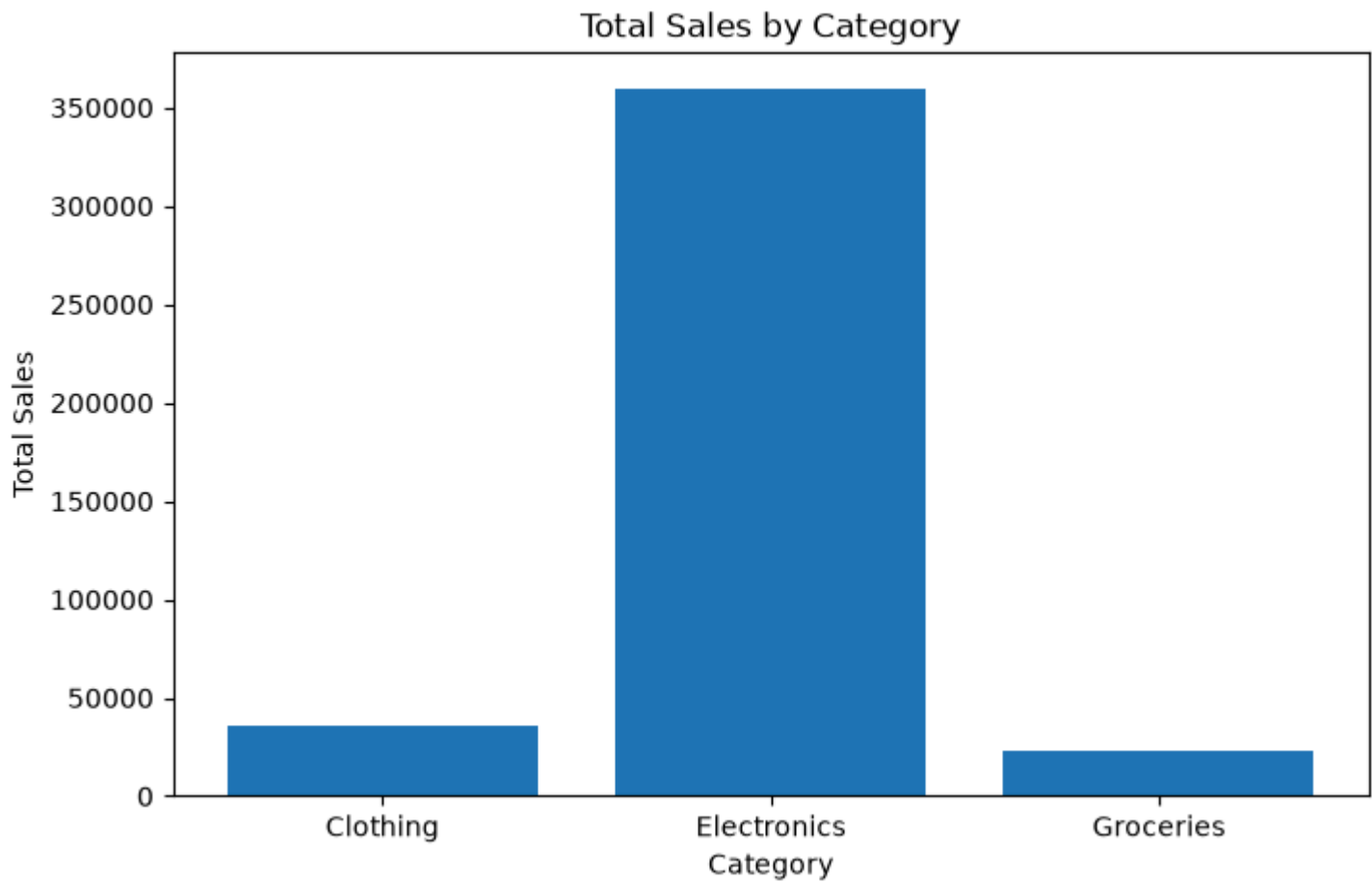
print(category_sales)
```

```
category
Clothing      35700
Electronics   359700
Groceries     23400
Name: sales_amount, dtype: int64
```

```
In [8]: plt.figure(figsize=(8, 5))

plt.bar(category_sales.index, category_sales.values)

plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")
plt.show()
```



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In [9]: #Task 3 - Monthly Sales Trend

sales_df["date"] = pd.to_datetime(sales_df["date"])
```

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In [10]: sales_df["month"] = sales_df["date"].dt.month
```

```
In [11]: monthly_sales = sales_df.groupby("month")["sales_amount"].sum()

print(monthly_sales)
```

```
month
1      58700
2      34000
3       9700
4      31300
5      64300
6      36400
7      10700
8      33700
9      61900
10     35400
11     10100
12     32600
Name: sales_amount, dtype: int64
```

```
In [13]: sales_df["month_name"] = sales_df["date"].dt.strftime("%b")

monthly_sales = sales_df.groupby(
    ["month", "month_name"]
)["sales_amount"].sum()

monthly_sales = monthly_sales.sort_index()

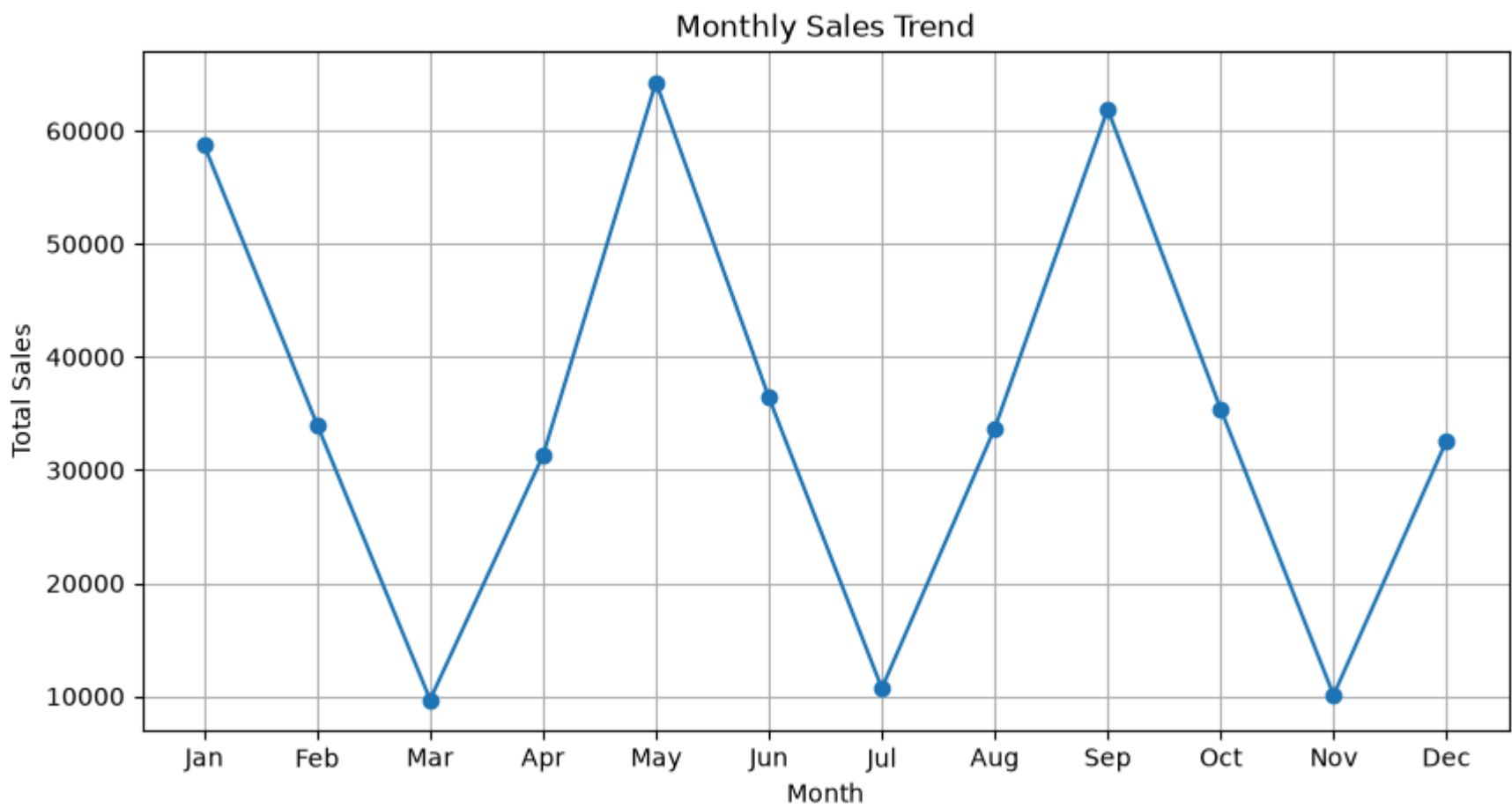
plt.figure(figsize=(10, 5))

plt.plot(
    monthly_sales.index.get_level_values("month_name"),
    monthly_sales.values,
    marker="o"
)

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")

plt.grid(True)

plt.show()
```



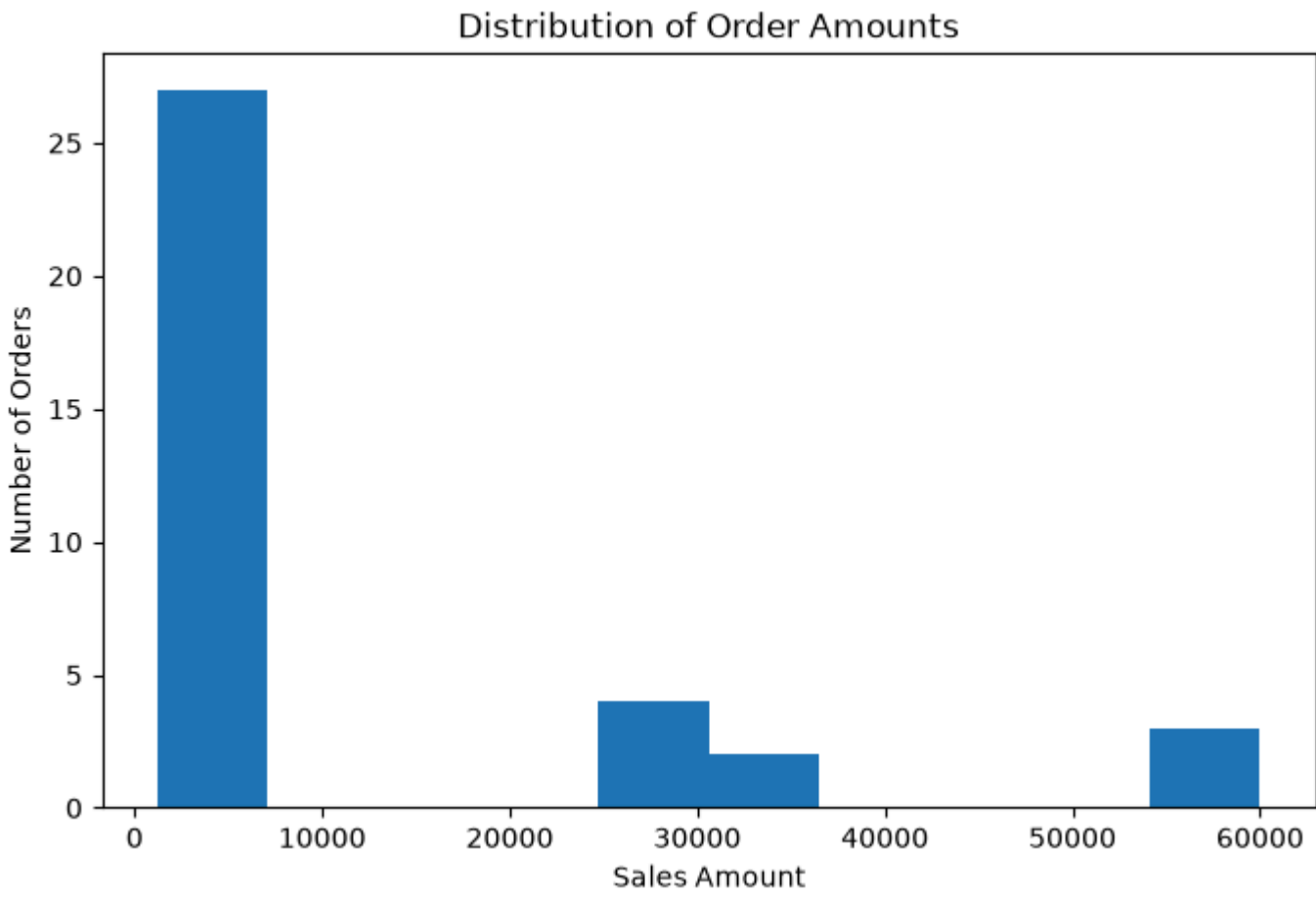
```
In [14]: #Task 4 - Histogram of Order Amounts

plt.figure(figsize=(8, 5))

plt.hist(sales_df["sales_amount"],bins=10)

plt.title("Distribution of Order Amounts")
plt.xlabel("Sales Amount")
plt.ylabel("Number of Orders")

plt.show()
```



In [15]: *#Task 5 - Annotate Category Sales and Export PNG*

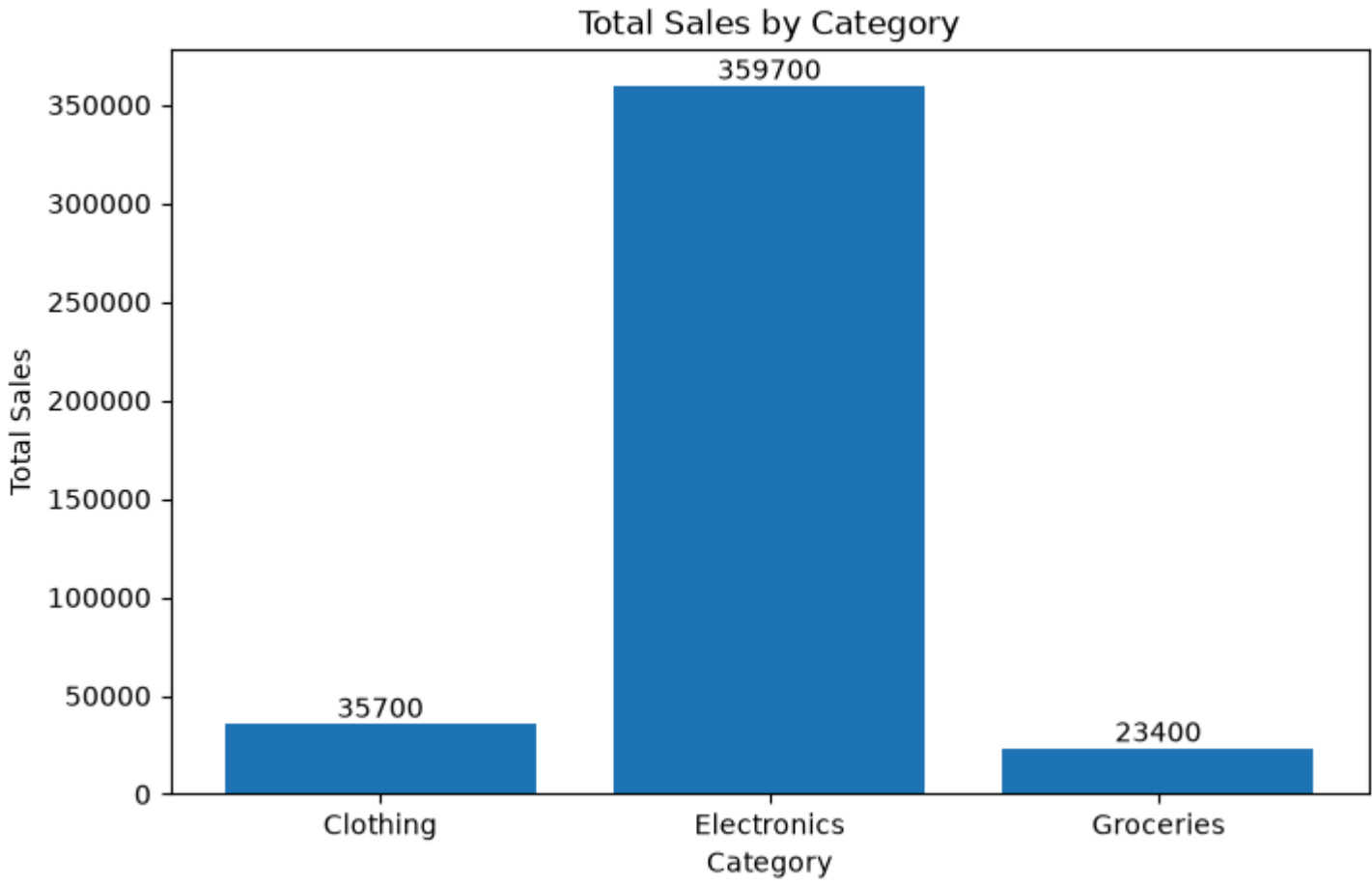
```
category_sales = sales_df.groupby("category")["sales_amount"].sum()
plt.figure(figsize=(8, 5))

bars = plt.bar(
    category_sales.index,
    category_sales.values
)

plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")

for bar in bars:
    height = bar.get_height()

    plt.text(
        bar.get_x() + bar.get_width() / 2,
        height,
        f"{height:.0f}",
        ha="center",
        va="bottom"
    )
```



```
plt.savefig(
    "category_sales_kpi.png",
    dpi=150,
    bbox_inches="tight"
)

plt.show()

<Figure size 640x480 with 0 Axes>
```

In []: